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COLLEGE OF COMPUTING, INFORMATICS AND MATHEMATIC

GROUP PROJECT
ISP688: INTELLIGENT DECISION-MAKING SUPPORT SYSTEM

TITLE: HEART DISEASE RISK PREDICTION

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1.0 INTRODUCTION

The Heart Disease Risk Prediction System is a data-driven and model driven Decision Support System (DSS) that predicts the risk of heart disease through the analysis of patient medical data. Age, gender, blood pressure, cholesterol level, ECG results, maximum heart rate, chest pain type, blood sugar reading and a target value indicating the presence or absence of heart disease are among the critical characteristics processed by the system. Using the data driven algorithm, the system offers an efficient and systematic way in order to identify patients who are at risk from early on. A user-friendly Power BI dashboard is then used to represent the output in order to assist medical professionals in properly understanding the results.

Early identification and timely detection of risk factors are critical because heart disease keeps on to be one of the main causes of death worldwide. Manual analysis of dispersed patient data is still used in many healthcare settings, which can be difficult and subject to errors. This system improves the efficiency and accuracy of decision making by combining all patient information into a single analytical system. Furthermore, using the Kaggle heart disease dataset will ensure that the system relies on accurate and real world medical data that is suitable for predictive modeling.

The overall goal of this research is to show how DSS technology and predictive analytics may be integrated to produce valuable insights that aid in clinical decision making. This system can help the healthcare professionals to improve early screening processes, and focus more on high risk patients more efficiently through user friendly visualizations and automated forecasting features.

1.1 Background

Millions of individuals worldwide suffer from heart disease, a serious public health concern which usually spreads silently until it reaches a critical stage. In order to figure out the risk, healthcare professionals usually analyze a variety of health signs, including maximal heart rate, ECG abnormalities, resting blood pressure, and cholesterol levels. Combining all of these factors provides important indicators, but manually evaluating each of them might be challenging, especially within crowded healthcare environments. Basic patient data, including age, gender, blood pressure, cholesterol level, ECG results,

kind of pain chest, blood sugar level, and the goal diagnostic showing the presence of heart disease, are included in the dataset used in this study.

Healthcare professionals could benefit especially from automated analysis that were possible by advances in artificial intelligence and machine learning. Predictive models can identify subtle changes in patient data that a human review could detect. In order to improve screening and diagnosis, numerous healthcare systems are starting to implement AI-driven technologies. By creating a DSS that uses predictive modelling to increase early detection accuracy, our research is in line with these current developments.

Since these public datasets, like the Kaggle Heart Disease Dataset are accessible, researchers and students may develop detailed prototypes that represent real applications in health care. The models for classifications can be developed based on the labeled, structured medical data included in these systems. In order to help healthcare professionals and healthcare organizations, we study on how these datasets may be transformed into useful decision support tools through this project.

1.1 Problem Statement

Heart attacks and other cardiovascular problems remain to be the major causes of death and illness worldwide. Many high risk patients do not receive early detection for heart disease because the screening process requires analysing several medical details manually (Li et al., 2024).. Even though healthcare continues to advance, several obstacles remain in place preventing precise detection of heart attack risk factors which include alcohol intake, smoking history, stress management, and factor genetics (Shuge et al., 2022). The likelihood of overlooking and misdiagnosing increases when the vital parameters for health such as blood sugar, blood pressure, heart rate, cholesterol rate, and ECG data are examined separately rather than all at once. This is especially common in situations when there are many patients and few medical professionals (Shesharao et al., 2024).

This challenge contributes to several serious consequences including late diagnosis, high risk patients not being identified in time, unnecessary delays in treatment, and increased treatment costs. Heart failure and myocardial infarction are examples of serious complications that usually develop from conditions that may be controlled or treated earlier (Hazzaa & Yildiz et al., 2025). Therefore, there is a clear need for a system that can support healthcare professionals by analysing patient data accurately and consistently. Current decision making tools have limitations in processing vast datasets which prevents healthcare professionals from generating actionable conclusions to develop personalized treatment strategies.

To address these issues, our group will develop a Decision Support System (DSS) that can analyse several patient details at once, predict the likelihood of heart disease, and identify high risk patients in order to manage these problems ((Hazzaa & Yildiz et al., 2025). The system will help healthcare professionals make decisions more quickly and accurately by providing clear visual outputs such as risk levels High, Medium, Low, are important variables that contribute, along with patient demographic patterns, and comparisons between healthy and heart disease patients. The system will show prediction results in a straightforward way for healthcare professionals using a Power BI dashboard. (Garg et., 2025).

1.2 Objectives

The objectives of these project are as below:

1. To develop a Decision Support System (DSS) for predicting heart disease risk based on medical data from the patient.
2. To develop a simple and user-friendly dashboard system that can help healthcare professionals visualise the risk level of patients and key medical indicators.
3. To minimize manual analysis and enhance the accuracy of patient screening with the help of model-driven decision support.
4. To support clinical decision making by classifying patients as low, moderate and high risk levels of heart disease.

1.3 User Requirement

The main users of the proposed Decision Support System are:

1. Doctors
2. Medical Staff
3. Hospital Management

Identifying user requirements is essential so that the DSS is designed based on actual clinical decision-making requirements. Doctors need clear and reliable patient risk categorisation to aid their screening, diagnosing and follow-up decisions, particularly when managing large numbers of patients. Medical staff benefit from simplified visual signalling that enables a quick interpretation of patient conditions without the need to analyse raw medical data. Hospital management, on the other hand, needs

aggregated statistics and trend analysis for strategic planning, policy formulation and efficient distribution of healthcare resources. By addressing these variety user requirements the proposed DSS will ensure usability, relevance and practical adoption within the healthcare setting.

1.4 Project Scope

In order to help patients to assess their health and living better lifestyles, the project aims to develop a data-driven and model driven decision support tool that predicts the risk of a heart attack. The scope includes data cleaning, model development, visualization of the outputs, and dashboard development using tools such as Excel, Power BI, or Python. Predictive models will be developed and validated using machine learning approaches to identify high risk individuals and groups. In order to support targeted public health initiatives, the system will generate dashboards and reports for healthcare professionals that provide information on patient risks, regional trends, and demographic trends.

The goal of this project is to create a DSS prototype that uses a machine learning classification model trained on the Kaggle Heart Disease Dataset, which includes the target variable which are age, gender, cholesterol level, resting blood pressure, maximum heart rate, chest pain type, fasting blood sugar, and electrocardiographic (ECG) results, to predict the risk of heart disease. To ensure the accuracy, practicality, and reliability, the models and system will be evaluated using historical data from the dataset.

The dashboard will display about risk categories, patients demographics, model prediction which is the outputs are limited to the related visual analytics and does not include any treatment recommendations, real time monitoring, and communication with healthcare information systems. The main purpose is to support screening decisions by generating predictions and visual summaries and demonstration on how such systems can be developed.

1.5 Project Significance

The main objective of this study is to reduce the chance of heart disease by looking at major cardiovascular factors and lifestyle aspects such as stress levels, intake of alcohol, and smoking. The outcomes from this research help healthcare professionals about how heart disease develops, which helps them develop targeted preventive strategies and public awareness campaigns for high risk individuals such as heavy drinkers and smokers. Healthcare professionals can make evidence-based, individualized

medical decisions that enhance patient outcomes while reducing the cost that comes with emergency care, long term recovery, and indirect economic burden by using advanced decision support systems (DSS) that have been applied from other domains.

The study improves both new data driven management techniques and heart attack preventive strategies with findings previously undetected patterns of cardiovascular health through thorough data analysis. In order to prevent severe issues and increase overall healthcare efficiency, the automated approach reduces human error, ensures consistent evaluation of patient data, and makes it easier to identify high risk individuals early on. This is useful in settings especially when healthcare professionals have a lot of patients to treat while facilities are limited.

Besides, this project shows how AI driven tools and DSS frameworks can improve current medical procedures and contribute to the developing area of digital healthcare. The technique enables easier, more accurate decision making, promotes additional technological advancement, and ultimately improves better patient care by presenting detailed medical information in straightforward, practical ways.

2.0 LITERATURE REVIEW

Heart disease continues to be one of the leading causes of death across the world, and early risk prediction is essential to mitigate fatal outcomes. However, diagnosis of heart disease is an intricate task because of the uncertainty, vagueness and interdependency of multiple clinical parameters such as age, blood pressure, cholesterol level and blood sugar. Traditional rule-based or statistical diagnostic methods often find it difficult to cope with the imprecise medical data and this has prompted researchers to consider the use of fuzzy logic based decision support systems for the prediction of risk for heart disease.

Fuzzy logic has been widely used in medical decision support systems since it closely resembles human reasoning and can be effectively used to deal with uncertain and ambiguous information. Unlike conventional Boolean logic which relies on strictly true or false values, fuzzy logic allows for partial membership values between 0 and 1 and thus is appropriate for clinical environments where patient symptoms do not always fall into clear cut categories (Kahtan et al., 2018). This characteristic allows fuzzy systems to model more accurately the medical decision making process in the real world.

Several research papers have shown the efficacy of fuzzy logic in diagnosis and prediction of heart disease. Kahtan et al. (2018) proposed a heart disease diagnosis using fuzzy logic by incorporating clinical parameters such as age, blood pressure, cholesterol and blood sugar. Their study demonstrated that fuzzy logic can help in reducing doctors work load with reliable diagnosis support especially in a situation where the relationship between the symptoms are unclear. The system was successful in translating numerical patient data into linguistic variables and using fuzzy rules to determine the risk of disease.

Similarly, Kowsigan et al. (2017) suggested a fuzzy logic-based prediction system for heart diseases by utilizing a small set of significant parameters. Their results showed that fuzzy logic systems have great potential to reach high accuracy of predictions and simplify the diagnostic process. The study has emphasized that fuzzy logic is effective for early detection and preventive healthcare for patients and healthcare providers to take the precautionary measures before the occurrence of severe symptoms.

Anooj (2011) further improved fuzzy logic-based decision support systems with weighted fuzzy rules for the heart disease risk prediction. By implementing different weights for clinical parameters, the provided system improved the prediction accuracy in comparison with conventional fuzzy systems. This approach proved that fuzzy logic not only devotes to decision-making under uncertainty but can also be improved to give more influence to risk factors in clinical diagnosis.

In addition, Arslankaya and Celik (2021) used fuzzy logic to predict the risk of heart attack by combining multiple clinical variables and rules defined by experts. Their research has confirmed fuzzy inference systems, primarily Mamdani that is for modelling complex medical conditions. The study mentioned the fact that fuzzy logic is suitable in healthcare applications because of its flexibility and ability to include expert knowledge.

Despite so much research work on fuzzy logic-based heart disease prediction, many of the existing systems focus mostly on standalone heart disease prediction models with not much of a combined function with user-friendly dashboards and decision support interfaces. Furthermore, the development of some systems is done as experimental tools without clear ideas on how to implement them in real-world healthcare environments. These gaps point to the need for a decision support system that does not only apply fuzzy logic for the accurate prediction of the risk of heart diseases, but also gives results through intuitive dashboards to assist healthcare professionals in their decision-making.

Therefore, in this project, a fuzzy logic based Decision Support System (DSS) to predict heart disease risk is proposed with the integration of clinical data processing, fuzzy inference and visualization with dashboard. By concentrating only on fuzzy logic techniques, the proposed system deals with uncertainty in the medical data and offers interpretable and actionable information to healthcare practitioners.

In summary, it is evident from the existing literature that model-driven DSS which involves the usage of fuzzy logic models are very effective in risk prediction of heart diseases. At the same time, data driven DSS facilitate in improving decision-making by analysing large scale clinical data and finding meaningful patterns and trends. These approaches aid in early detection, help increase the accuracy of diagnoses and help healthcare professionals make informed decisions. Based on these findings, in this project, a data driven and model driven DSS approach for heart disease risk prediction is adopted, where the focus lies on the analytical modeling and predictive insights towards clinical decision making processes.

3.0 METHODOLOGY

This section outlines the research methodology used in developing a model-driven decision support system (DSS) to analyse heart disease risk. The methodology utilises the data-based, quantitative strategy to study the associations between demographic factors, lifestyle habits, and medical signs that predispose individuals to heart diseases. The system enhances the decision making process and detecting risks early on by combining predictive models with data visualisation tools.

The research involves secondary data based on an open-source dataset and adds statistical and machine learning models to derive predictive information. The methodology is structured into several stages, such as preliminary research, data gathering, data interpretation with the help of a data dictionary, model-driven analysis, system usability testing, and system creation. This systematic methodology can make the proposed decision support system reliable, interpretable, and usable.

The overall workflow of the proposed DSS is based on a structured and systematic process. It starts with the data acquisition and data preprocessing to ensure the quality of the data and then risk prediction with the help of analytical models and machine learning models. The predicted outcomes are then visualised in the form of interactive dashboards where these are able to visually interpret the risk levels for healthcare professionals in an efficient way. Finally, these insights support the clinical screening, decision making on prioritisation of patients. This end-to-end workflow ensures that the DSS helps to transform raw medical data into actionable information which will help healthcare decisions directly.

3.1 Research Methodology Design

The research design is model-driven DSS research design where the core element in decision making is the analytical models. This methodology has several stages as follows: preliminary study, data collection, data dictionary preparation, model development, system usability analysis and system development. Each step will help in the process of transforming raw data to valuable insights.

The model-driven methodology is aimed at building predictive models to estimate the risk of heart disease using a variety of input variables. The results of these models are built into an interactive dashboard where the users can browse predictions and comprehend risk factors.

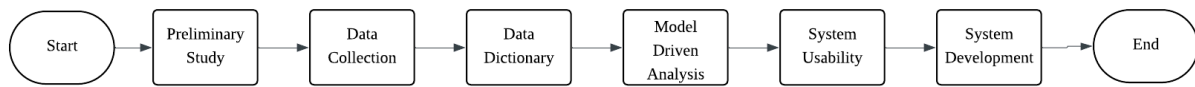


Figure 3.1 Research Methodology Phase

3.2 Preliminary Study

The initial research aims at knowing the problem area across the heart disease problem and finding out some of the significant issues leading to heart attack risk. The step will entail examination of the literature review and available healthcare report to establish common medical indicators related to cardiovascular diseases. The factors like age, gender, blood pressure, cholesterol level, blood sugar level, maximum heart rate, type of chest pains, and electrocardiographic (ECG), results were found to be important contributors to the risk of heart diseases and align with the characteristics that were present in the data chosen.

Moreover, this step will assess the appropriateness of the implementation of a model-driven Decision Support System (DSS) model to predict the risk of heart diseases. The results of the initial research justify the adoption of predictive models to process patient data and produce risk categories that can help medical staff to screen patients early and make decisions. The findings of this phase were used to select dataset attributes and the general design of the proposed system.

3.3 Data Acquisition

To collect data, the dataset to be used in the study was found in the Kaggle website at

<https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset/data>

The dataset includes structured medical records in relation to heart disease and it gives information on several patient attributes like age, gender, blood pressure, cholesterol-level, chest pain type, maximum heart rate, and presence or absence of heart disease. The collected data are used as the main input for analysis and shown in Figure 3.2.

1	Age	Chest_Pain_Type	Cholesterol_Level	Exercise_Induced	Fasting_Blood_Sugar	Heart_Disease	Major_Vessels_Count	Max_Heart_Rate	Resting_Blood_Pressure	Resting_ECG	Sex
2	29	1	204	0	0	1	0	202	130	0	1
3	34	1	210	0	0	1	0	192	118	1	0
4	34	3	182	0	0	1	0	174	118	0	1
5	35	0	183	0	0	1	0	182	138	1	0
6	35	0	198	1	0	0	0	130	120	1	1
7	35	0	282	1	0	0	0	156	126	0	1
8	35	1	192	0	0	1	0	174	122	1	1
9	37	2	215	0	0	1	0	170	120	1	0
10	37	2	250	0	0	1	0	187	130	1	1
11	38	2	175	0	0	1	4	173	138	1	1
12	38	3	231	1	0	0	0	182	120	1	1
13	39	0	219	0	0	0	0	140	118	1	1
14	39	2	199	0	0	1	0	179	94	1	0
15	39	2	220	0	0	1	0	152	138	1	0
16	39	2	321	0	0	1	0	182	140	0	1
17	40	0	167	1	0	0	0	114	110	0	1
18	40	0	223	0	0	0	0	181	152	1	1
19	40	3	199	1	0	1	0	178	140	1	1

Figure 3.2: Sample records from the Heart Disease dataset

3.4 Data Dictionary

Table 3.1: Data dictionary of patient attributes and datatype

Attributes	Description	Datatype
Age	age in years	Whole number
Chest_Pain_Type	chest pain type (4 values)	Whole number
Choleterol_Level	serum cholesterol in mg/dl	Whole number
Exercise_Induced Angina	exercise induced angina (1=yes; 0=no)	Whole number
Fasting_Blood_Sugar	(fasting blood sugar & gt; 120 mg/dl) (1=true; 0=false)	Whole number
Heart_Disease	presence of heart disease (1=yes, 0=no)	Whole number
Major_Vessels_Count	number of major vessels (0-3) colored by fluoroscopy	Whole number
Max_Heart_Rate	maximum heart rate achieved	Whole number
Resting_Blood_Pressure	resting blood pressure (in mm Hg on admission to the hospital) (values 0,1,2)	Whole number
Resting_ECG	Resting electrocardiographic results	Whole number
Sex	(1=male; 0= female)	Whole number
ST_Depression	ST depression induced by exercise relative to rest	Decimal number
ST_Slope	the slope of the peak exercise ST segment	Whole number
Thalassemia	Thalassemia condition (thal: 0=normal; 1=fixed defect; 2=reversible effect)	Whole number

3.5 System Usability

The usability of the developed Power BI dashboard is considered in the scope of layout design, accessibility, interactivity, and the overall ability to support decision-making. The dashboard enables the user to dive into the trend of heart disease using several health indicators, including age, gender, chest pains type, disease status, and thalassemia category. The slicers and filters can also be interactive allowing users to easily narrow down on specific groups of patients with the visuals being arranged in a well structured way to increase readability and interpretation. The charts utilized such as line charts, donut charts and treemaps are a visual representation of information that requires less cognitive effort and aids in rapid analysis. In general, it is evident that the dashboard is accessible, convenient, and offers an effective analytical space to learn more about the distribution of heart disease among various medical variables.

3.5.1 Cards



Figure 3.3: Cards layout in dashboard

The top of the dashboard contains the KPI cards that will give a brief summary of the most significant metrics of heart disease risk screening. Such indicators enable healthcare specialists to get a clear insight into the general situation in the dataset and the scope of the high-risk clients first before delving into the specific visualisations. The cards are useful for advancing situational awareness and also help clinicians make better and faster screening decisions by presenting the key figures in a summarised form.

The initial card, Total Patient, shows the total number of patient records in the dataset which forms the point of reference in relation to the whole population of patients. The second card, High Risk Patient, displays the amount of the patients who have been classified as high-risk according to the risk scoring method used. The number is necessary in order to prioritize patients, who might need urgent care and additional diagnostic processes. The third card, Heart Disease Cases, demonstrates the amount of confirmed cases of heart disease ($\text{Heart_Disease} = 1$), which is the actual outcome variable that will be utilized in comparing and validating in the dashboard. Finally, there is the Average Risk Score (High Risk) card which illustrates mean risk of the high-risk group and serves as yet another severity indicator representing the entire level of risk intensity in this priority group.

On the whole, these KPI cards serve as fast-access indicators helping the clinical community to make a decision in the form of summarising the population of patients, the burden of high risks, confirmed cases, and the level of severity of high-risk patients and then users can examine relationships across risk categories, symptoms, and demographic patterns on the rest of the dashboard.

3.5.2 Cluster Column Chart

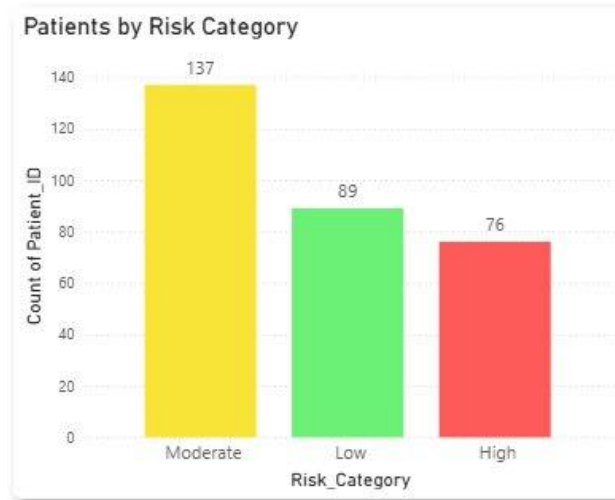


Figure 3.4: Cluster Column chart layout in the dashboard

The Patients by Risk Category clustered column chart shows how the patients are distributed within the three risk groups produced by the risk scoring model: Low, Moderate and High. The risk category is represented as x-axis and the number of patients in each category as y-axis. This visualization can be used in screening to diagnose patients at an early stage since it enables the respective medical personnel to establish the percentage of patients that need varying amounts of clinical care very fast.

As the chart shows, there are the largest number of patients in the Moderate-risk group, the next number in the Low-risk group, and the final group, the High-risk group, is a small percentage of the population. This distribution gives a background of the general risk profile of the database and assists in prioritization before additional screening or follow-up evaluation is performed. The color coding (Low, yellow, High) contributes to making the information easier to read and to the common sense of how severe the risk is (the clinical interpretation of the risk severity).

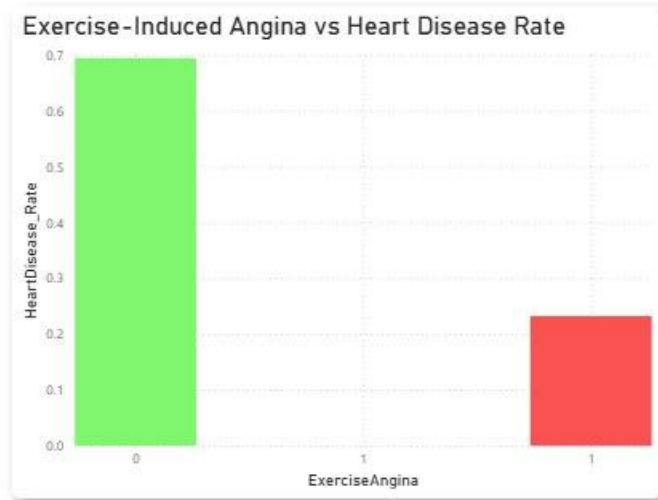


Figure 3.5: Cluster Column chart layout in the dashboard

Exercise-Induced Angina vs Heart Disease Rate clustered column chart compares the heart disease rate among the patients who experience exercise induced angina and those who do not. The x-axis is used to indicate a binary angina (0 = No, 1 = Yes), and the y-axis indicates the rate of heart disease which is the proportion of patients with heart disease in each group.

This chart offers clinical insight based on the symptoms as it gives priority on the relationship that exists between exercise-induced angina and occurrence of heart diseases. An increased incidence of the heart disease in the group with angina brought about by physical activity indicates that the symptom can be a powerful clinical marker and can help a health expert diagnose patients who need further cardiovascular assessment. In general, this visualization enhances the decision support role of the dashboard by associating symptom evidence with prevalence of heart diseases.

3.5.3 Cluster Bar Chart

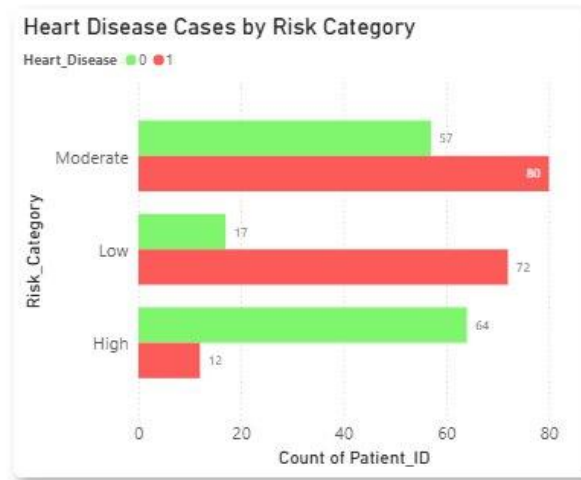


Figure 3.6: Cluster Bar Chart layout in the dashboard

The Heart Disease Cases by Risk Category clustered bar chart is used to compare the actual heart disease outcomes of the three risk categories projected to happen (Low, Moderate, High). The y-axis indicates the risk category whereas the x-axis indicates the number of patients. The Heart_Disease label (0 = no heart disease, 1 = heart disease) divides the bars allowing the users to explore the quality of the risk classification in terms of cases of confirmed heart diseases.

This visualization contributes to the legitimacy of the logic of risk prediction by displaying the fact that the categories of higher risk are associated with more cases of heart disease. The ideal situation is that the High-risk group must have the higher number of Heart_Disease = 1 cases. Thus, the chart assists medical workers as it will produce evidence that the risk stratification is a clinically significant one which may be trusted to make early screening decisions.

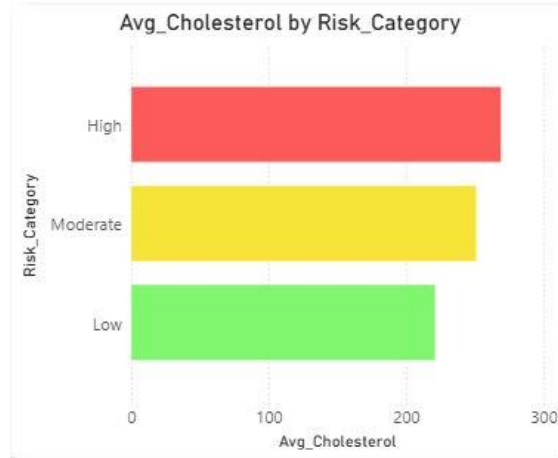


Figure 3.7: Cluster Bar Chart layout in the dashboard

The bar chart of the Average Cholesterol by Risk Category indicates the mean cholesterol of the patients in each of the risk groups (Low, Moderate, High). Risk categories are presented in the y-axis, whereas the calculated average cholesterol value is presented in the x-axis. This chart gives a clinical explanation of the risk stratification which can be interpreted by displaying the variation in cholesterol level in the varying risk levels.

The visualization denotes that the High-risk group bears the highest mean cholesterol level and then the Moderate-risk group then the Low-risk group displays the lowest mean cholesterol. This trend is embracing clinical knowledge because high cholesterol is one of the known factors that contribute to cardiovascular disease. Consequently, the chart empowers the dashboard with a decision-support role, which is connecting the projected risk types to a critical physiological indicator.

3.5.4 Stacked Bar Chart

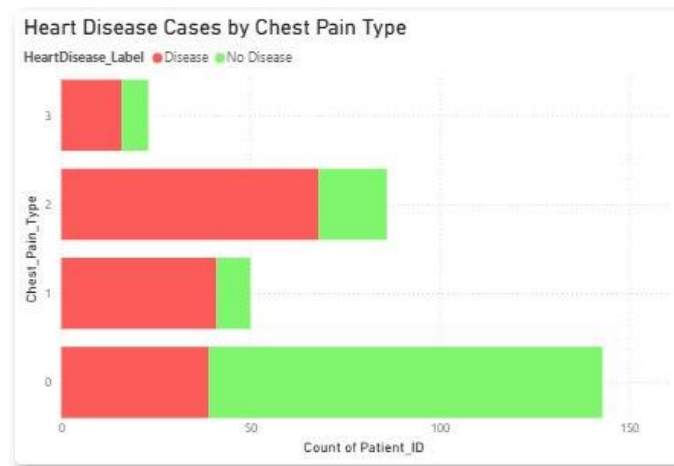


Figure 3.8: Stacked Bar chart layout in the dashboard

The Heart Disease Cases by Chest Pain Type stacked bar chart shows the relationship between different types of chest pains and prevalence of heart disease. The y-axis shows the categories of ChestPainType, whereas the x-axis shows the number of records of patients. All the bars are divided in two depending on the outcome label of heart disease Disease (Heart Disease = 1) and No Disease (Heart Disease = 0). This enables the users to compare the ratio of cases of heart diseases in the groups of chest pains.

This graphic is clinically informative since the type of chest pain is one of the major symptoms employed in screening cardiovascular. Through comparing disease and non-disease rates of the two types of chest pains, health care providers can determine the patterns of chest pains to be more closely linked to heart disease. Thus, the chart makes it easy to detect the disease early by connecting indicators of symptoms that appear to the disease and supplement the analysis of risk categories in the dashboard.

3.5.5 Scatter Plot

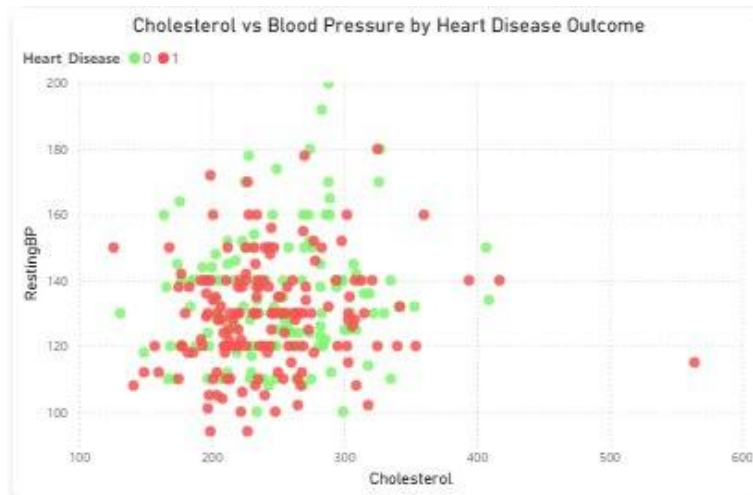


Figure 3.9: Scatter plot chart layout in the dashboard

The scatter plot represents the correlation between the cholesterol level and the resting blood pressure (BP) and each point has got the record of a patient. The x-axis indicates the cholesterol levels whereas the y-axis indicates the blood pressure levels. The points are colour coded according to the outcome of heart disease (Heart Disease = 0 no disease, Heart Disease = 1 disease), and thus allow healthcare professionals to compare the clinical patterns of patients with and without heart disease.

The visualisation will aid in decision making because it shows that heart disease risk is a combination of various contributing factors and not just cholesterol level. The distributions of the points demonstrate that not all high cholesterol patients have to have heart disease and that not all the heart disease patients are also high cholesterol disease at moderate cholesterol levels depending on their blood pressure. Hence, the decision support aspect of the dashboard can be enhanced by the scatter plot because they offer a more explicit multi-factor clinical interpretation of the early screening and patient prioritisation.

3.5.6 Line Chart

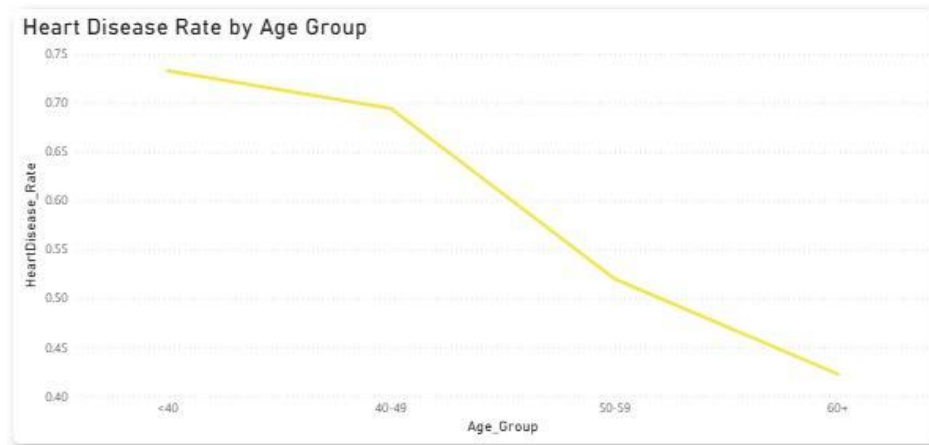


Figure 3.10: Line Chart layout in the dashboard

The Heart Disease Rate by Age Group line chart illustrates the data trend of the prevalence of heart disease among various categories of people. The x-axis is the Age_Group, (<40, 40-49, 50-59, 60+), and the y-axis is the Heart Disease Rate, which is the percentage of patients with heart disease in each Age Group. This line graph illustrates the changes in the likelihood of heart disease due to old age.

This chart will assist in demographic-based screening by helping medical workers to understand the age groups that are at the risk of higher prevalence of heart diseases. As age is one of the primary risk factors in the cardiovascular condition, tracking the trend among the different age groups can be of help when prioritizing early screening and following up of the vulnerable population. Within the dashboard, this type of visualization is the complement to the symptom and risk category charts that introduces a population-level demographic picture to the risk prediction of heart disease.

3.5.7 Table

Patient ID	Age	Gender	Chest Pain Type	ExerciseAngina	Risk Score	Risk Category
22	55	Female		0	1	10 High
23	56	Female		0	1	10 High
26	59	Male		0	1	10 High
27	60	Male		0	1	10 High
37	70	Male		2	1	10 High
18	51	Male		0	1	9 High
22	55	Male		0	1	9 High
24	57	Male		0	1	9 High
25	58	Female		0	1	9 High
28	61	Female		0	1	9 High
28	61	Male		2	1	9 High
29	62	Female		0	1	9 High
34	67	Male		0	1	9 High
35	68	Male		2	1	9 High
13	46	Male		0	1	8 High
18	51	Male		0	1	8 High
20	53	Male		0	1	8 High
22	55	Male		0	1	8 High
23	56	Female		0	1	8 High

Figure 3.11: Table layout in the dashboard

The High-Risk Patients List table in the dashboard is an easy to understand and well-structured table that gives a clear picture of the patients that have been found to have a high chance of heart disease. It displays the key details of the patient like his or her Patient ID, Age, Gender, the type of Chest Pain, Exercise-Induced Angina, and calculated Risk Score with the Risk Category. With such attributes shown in a single table, the users can easily assess and compare the profiles of the patients without searching through many charts or pages.

The table assists in clinical decision-making since the medical staff can filter, sort, and narrow to patients who have the highest risk scores, so it becomes easier to prioritize further screening, monitoring, or early intervention. It is also the detailed supporting element to the overall dashboard where the user is able to validate the risk patterns by the individual patient level and what patient factors result into being classified as high-risk.

3.6 System Development

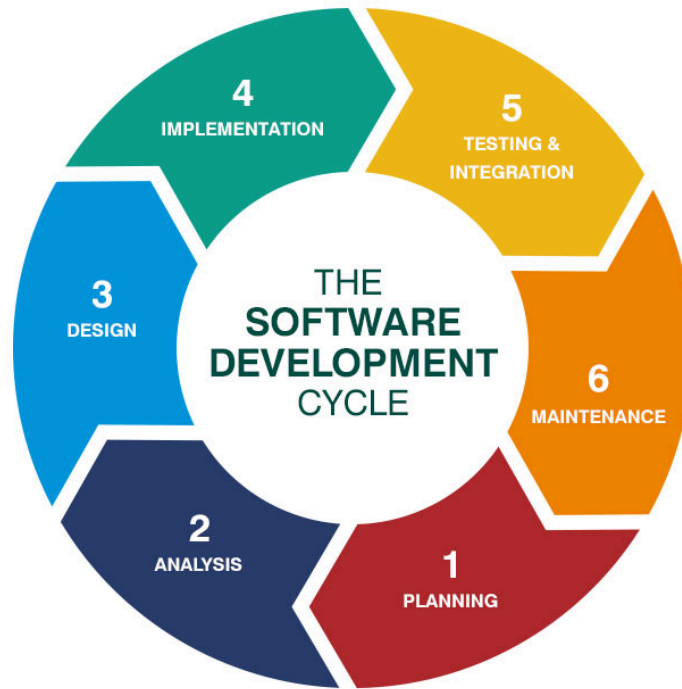


Figure 3.12: The development cycle for the DSS

In this phase of the development process of the Decision Support System (DSS) is described according to the software development lifecycle. The system has been developed incrementally to ensure the accuracy, usability and reliability of the final dashboard. The development cycle is divided into some of the most important phases, starting from planning and analyzing the requirements, and then we move on to the phase of designing the dashboard where data structure, visual alignment, and metrics were defined. The implementation phase includes the development of the Power BI dashboard and the integration of the datasets and later the testing and validation phase to ensure proper functionality and meaningful insights. Finally, maintenance and refinement are done to provide both improved performance and improved decision-making support to users.

3.7 Implementation Strategy

The implementation strategy of the proposed Decision Support System (DSS) is designed to ensure that the system can be successfully developed and utilised to the real healthcare environments. The strategy is made up of two main phases, which are system development and system usage. During the development of the system, medical data of patients is gathered from the existing datasets in healthcare and passes

through data cleaning and preprocessing steps, in order to achieve accuracy and consistency. Analytical and predictive models are then used to determine the risk of heart disease and group patients into various levels of risk. The processed results are then integrated into a Power BI dashboard, which is then the primary user interface for the DSS. In terms of system usage, healthcare professionals use the dashboard to review the risk levels of the patients, identify high-risk individuals and prioritise them for further screening and medical intervention, and aggregated insights for managerial planning and resource allocation.

This implementation strategy makes sure that technical processes are directly transformed into practical decision support. By organising the system from data acquisition up to visualization on a dashboard, the DSS helps to bridge the gap between the complex data analytical output and the clinical decision making need. The combining of predictive models and interactive dashboard has allowed healthcare professionals to easily interpret results without necessarily knowing how to use advancements. As a result, the system supports more effective decision-making, decreases reliance on manual data analysis and improves the overall effectiveness of patient screening and prioritisation in high workload healthcare settings.

In addition to the use of conventional predictive and analytical models, a strategy based on fuzzy logic is implemented for the DSS, to increase the degree of interpretability and decision support. Fuzzy logic enables risk for patients to be modeled in terms of linguistic variables such as low, moderate, and high risks, and this is very close to the way healthcare professionals reason in real-world clinical situations. This is especially useful when there is uncertainty and imprecision in medical data such as cholesterol borderlines or blood pressure. Within the interface fuzzy logic results are represented by clear risk categories, colour-coded visualisations and dashboards with indicators to make difficult to understand numerical results meaningful and actionable. By incorporating fuzzy logic into the interface design, the DSS creates a more transparent, user-friendly, and trustworthy system for its recommendations, reinforcing its position as a clinical decision support tool and not just a clinical decision analysis system.

3.8 System Functionality

The proposed Heart Disease Risk Prediction system is a computerized Decision Support System (DSS) that is designed to help healthcare professionals assess patient risk and aid in preventive clinical decision making. The system combines fuzzy logic modelling with interactive dashboard visualization in order to convert medical data into actionable insights.

a. Functionality Market Area

The system functions under the healthcare domain, more specifically, medical decision support and preventive healthcare management. Its major functional focus is on clinical screening as well as the diagnosis assistance and risk assessment in relation to heart disease. This DSS differs from systems that are in marketing or manufacturing because it is suited to the healthcare setting where decisions must be made in the context of uncertainty, patient safety, and timely intervention. The system supports clinical operations as well as healthcare management by enabling early identification of high-risk patients and support for planning for preventive care.

b. Type of Computerized Tool

The proposed system is classified under Decision Support System (DSS) having integrated Business Intelligence (BI) features.

- As a Decision Support System, it helps healthcare professionals to make an informed decision by analysing patient data, applying fuzzy logic rules and categorising the risk level of heart disease.
- As a Business Intelligence tool, the system includes dashboards, charts, and key performance indicators (KPIs) to display results of the analysis in a visual and interactive way.

c. Type of Data Processed by the System

The system processes structured healthcare data which includes:

- Demographic information, such as age and gender.
- Clinical measurements such as blood pressure, cholesterol level, blood sugar and heart rate indicators.
- Lifestyle and medical condition indicators such as smoking, obesity, and diabetes
- Symptom related data like chest pain

These data types are of numerical and categorical nature. The system is based on the conversion of crisp numerical values into fuzzy linguistic variables (e.g. low, medium, high) in order to support reasoning under uncertainty. The processed data is then analysed based on fuzzy inference rules to produce risk classification for patients.

d. Utilisation of Web Technologies

Web technologies are utilised mainly via the dashboard of the system and interface layer. The Power BI dashboard is the web-based visual interface through which the user can interact with the system through filtering, selection, and drill down features. These web-based capabilities allow healthcare professionals to access the results of risk analysis in a dynamic way and interpret data without requiring the actual interaction with the underlying models.

Additionally, use of web-based visualization technologies enables scalability and accessibility to support the deployment of the DSS in different healthcare environments. Although the system is not dependent on complex web services and APIs, the integration of web-enabled dashboards improves usability, real-time interaction and collaborative decision-making.

In summary, the proposed system is a healthcare-focused Decision Support System which combines fuzzy logic modelling and business intelligence visualization. It processes structured medical data, works in the preventive healthcare domain, and uses web-based technologies in the form of dashboard analysis to support accurate, interpretable and efficient decision-making relating to prediction of heart disease risk.

4.0 INTERFACE



Figure 4.1: The Interface dashboard in PowerBi

The interface of the Heart Disease Risk Prediction Decision Support System (DSS) was developed with a high focus on readability, usability, and accessibility to make sure that medical workers can perceive the predictive results easily. The dashboard was created using Power BI, and it offers a well-structured and interactive system that allows users to navigate the patient data and categories of risks, as well as forecasts, without high-technology skills.

The Heart Disease Risk Prediction dashboard is created to help in clinical decision making by the use of a clear and structured analytical storyline that demonstrates how medical practitioners evaluate risk of a patient in practice. As soon as users enter the dashboard, the top cards of KPIs will give them a general picture of the present situation by providing the number of patients, patients with high risks, and the number of confirmed heart diseases, and the average risk score, which will allow them to quickly understand the situation before exploring it in greater detail. The dashboard continues with patient risk categorisation in which people are grouped into low, moderate and high risk groups using predictive risk scores, facilitating early screening and prioritisation. It is followed by a validation phase where the categories of risks as predicted are compared with the actual outcome of heart diseases, so that the clinicians can determine the reliability and clinical relevance of the risk prediction. To provide the underlying rationale behind the classification of high-risk, there are factor-based visualisations of average cholesterol by risk category, which connects the medical indicators with the risk outcomes. Probably, other symptom-oriented charts such as exercise-induced angina and the type of chest pain will also contribute to clinical knowledge by showing the relationship between the appearance of certain symptoms

and the occurrence of heart disease. Gender by age group demographic analyses at the time will give population level data to determine the vulnerable groups with an increased prevalence of the disease. Lastly, the dashboard ends with a comprehensive list of high-risk patients, which helps to convert analytical findings into practical decisions because the healthcare professionals can now prioritize the patients, who are to receive further screening, monitoring, and early intervention.

4.1 Layout Design

The dashboard is organized so as to give a clear view of the patient risk data. At the top, KPI cards provide brief information about the main statistics of the total patients, high-risk patients, confirmed cases of heart diseases, and average risk scores. Under these several visualizations are set in logical order to move the user through the general trends of the population back to specific risk factors analysis. Bar charts that are clustered, scatter graphs, line graphs, and categorical comparisons (e.g., cholesterol vs. blood pressure, rate of heart disease by age group, type of chest pain and gender) are all included in charts. The use of color codification (green-low-risk, yellow- moderate, red- high) improves the readability factor and enables quick recognition of categories of patients.

4.2 Interactivity

The dashboard is focused on visual simplicity and easy navigation as opposed to multifaceted filtering. All the charts are intended to reflect the correlations between medical characteristics and heart disease outcomes. Results are readable through the visualizations and users can get to the specific risk categories through the general statistics. The availability of a list of high-risk patients also helps to access patient information in more detail without the need to do any additional filtering.

4.3 Accessibility

The interface will be user friendly and easy to use so that the medical staff with diverse degrees of technical skills can use the system successfully. Labeling the charts, legends, and metrics are clearly defined avoiding ambiguity and easy understanding. The dashboard is able to be installed on a variety of devices, such as desktops and tablets, which makes it fit a variety of healthcare environments.

4.4 Decision Support

The proposed Heart Disease Risk Prediction Decision Support System (DSS) is aimed at supporting multi-levels of healthcare decision making by converting complex patient data into understandable, interpretable and actionable insights. The system helps healthcare professionals to evaluate patient risk, to prioritise clinical actions and to plan preventive healthcare strategies.

Decision Supported by the System

At the clinical level, the DSS assists the decision making related to patient risk assessment and prioritisation. By dividing patients into low, moderate and high risk groups for heart disease, the system helps doctors to decide which patients need immediate screening, further diagnostic tests or preventive intervention. This helps with the need for manual judgement in dealing with large amounts of patient data.

At the organisational level, the DSS is facilitating resource allocation and preventive planning decisions. Aggregated risk statistics enable hospital management to detect trends in heart disease risk in various age groups, gender or lifestyle factors that can then be used to make informed decisions about screening programmes, staffing and distribution of healthcare resources.

Software Featured Supporting Decision Making

The system has a number of key software features that make it more powerful in decision support:

1. Fuzzy Logic Modelling

The system utilizes fuzzy logic as a way of treating the uncertain data in the medical field, so that patient conditions can be assessed based on such linguistic terms as low, medium and high risk. This approach is very similar to human clinical reasoning and makes the interpretation even better.

2. Fuzzy Rule Base and Inference Method

Expert-defined IF-THEN rules are applied by using Mamdani fuzzy inference method, this allows the system to analyse complex relationships between multiple clinical indicators at the same time.

3. Interactive Dashboard Visualization

The Power BI Dashboard displays risk levels, trends and summary indicators in charts and KPI cards, enabling users to easily interpret the findings and identify high-risk patients.

4. Filtering and Drill down capabilities

Users can filter patient data based on demographic or clinical attributes and drill down to the patient level supporting both overview analysis and for detailed decision making.

Level of Managerial Support

The DSS is a decision support capability at a number of different managerial levels:

- **Operational Level**
Supports day-to-day clinical decisions e.g. for identifying high-risk patients for immediate screening or follow-up.
- **Tactical Level**
Helps in department level planning by analysing the distribution of patient's risk and identifying the priority groups for the targeted screening programmes.
- **Strategic Level**
Aids in long-term planning decisions by aggregating insights on trends in heart disease that can be used to inform preventive healthcare policies and strategies on how to invest resources

Individual and Group Decision Making

The system provides for individual as well as group decision making. For individual decision making, the doctors can use patient specific risk results to guide clinical judgement in consultations. For group decision making, aggregated dashboards help multidisciplinary healthcare teams and management staff to evaluate trends, discuss preventive strategies and healthcare intervention together.

Suggestion of Alternatives

The DSS gives decision alternatives in the form of risk-based categorisation and comparative analysis instead of a fixed single recommendation. For example, high risk patients can be offered immediate diagnostic testing, lifestyle intervention programmes or specialist referral and moderate risk patients can be offered monitoring and follow-up screening. This gives healthcare professionals the ability to consider several potential actions and the most appropriate one in a given context.

5.0 SYSTEM BENEFITS

The proposed Heart Disease Risk Prediction Decision Support System (DSS) offers several important benefits to healthcare professionals and healthcare organisations. One of the major advantages is the early detection of high-risk patients for heart diseases, which leads to early screening and intervention in a preventive manner. The system also eliminates some of the manual work involved in analysing multiple patient indicators by automating risk assessment and providing results in a structured and visual format. In addition, the DSS assists in quicker and more accurate clinical decision-making, categorising patients into different risk levels as healthcare professionals can effectively prioritise patients based on severity of risk.

These benefits directly address the important challenges experienced in the healthcare environment, especially large numbers of patients and shortages of medical personnel. By turning complex patient data into actionable data identification via predictive models and interactive dashboards, DSS helps with building situational awareness and less cognitive load during professional clinical decision making. Early identification of high-risk patients helps prevent the progression of the disease and minimise any chances of late diagnosis which can lead to high cost of treatment and severe health outcomes. Furthermore, the structured categorisation of risk and the visual analytics enhance consistency in patient evaluation to ensure that screening decisions are evidence-based, transparent and reproducible. As a result, the proposed DSS plays a role in better healthcare efficiency, improved patient outcomes, and more effective medical resources allocation.

Overall, the system benefits show the practical value of incorporating decision support systems into preventive healthcare and clinical screening processes.

6.0 PROCESS AND WORKFLOW

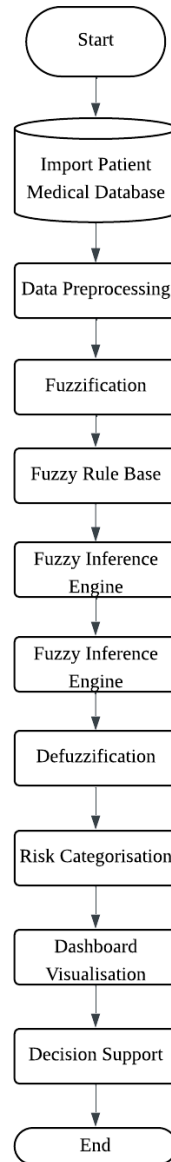


Figure 6.1: The workflow of proposed fuzzy logic based Decision Support System (DSS) for heart disease risk prediction

The proposed Heart Disease Risk Prediction Decision Support System (DSS) follows a structured and systematic process that aims at transforming raw patient data into actionable clinical insights. The workflow is organised into sequential stages in order to guarantee data accuracy, reliable risks evaluation and effective decision support for healthcare professionals.

The process starts with data acquisition, in which medical data of the patient including age, blood pressure, cholesterol level, blood sugar, and other relevant clinical indicators are gathered from already available data from healthcare institutions. This data is the main input for the DSS. In order to guarantee the quality and reliability of the system output, the collected data goes through a data preprocessing stage, involving data cleaning, validation, and standardisation. This step is used to eliminate inconsistency and prepare the data for further analysis.

Next is the preprocessed data that is fed to the fuzzy logic inference process which is the core of the DSS. In this stage, crisp values of inputs are transformed into fuzzy linguistic variables using predefined membership functions by a process known as fuzzification. These linguistic variables stand for clinical concepts such as low, medium and high risk levels. The fuzzy rule base developed according to expert knowledge and medical guidelines is then used to assess relationships between the input parameters and the risk of heart disease. The fuzzy inference engine uses these rules to produce fuzzy output values which are then converted into exact values of risk through the process of defuzzification.

Following risk assessment the results are incorporated into the decision support and visualisation stage. The system presents patient risk levels in the form of an interactive dashboard that includes risk categories, trends and summary indicators. This simplifies healthcare professionals from having to interpret the results quickly without analysing any complex numerical data. The dashboard supports filtering, comparison, and drill down at the aggregated and patient level respectively, giving the user's access to explore the data on the aggregated or patient level.

Finally, the DSS provides support for the clinical decision-making process and integration of workflows. Healthcare professionals take the outputs of the risk categorisation, which is used to prioritise high-risk patients for further screening, diagnostic tests, or medical intervention. At the organisational level, the aggregated insights-bringing organisations together in the planning, monitoring and allocation of resources. This end to end workflow is the key guarantee that the DSS can not only engage in risk prediction towards heart diseases, it is also able to contribute in a successful preventive healthcare and evidence-based decision making process.

7.0 FUTURE WORK

The existing prototype of Heart Disease Risk Prediction Decision Support System (DSS) proves the idea of applying predictive analytics and interactive dashboards to assist healthcare professionals during the process of early screening. But, the system can be developed in a few aspects and improved in order to enhance its accuracy, usability, and practical application in the real world. The future work can be divided into technical improvement and clinical integration and the wider healthcare impact.

7.1 Technical Enhancements

- Addition of Dataset Sources.

Although the Kaggle dataset has a solid background, it is necessary to consider larger and more differentiated datasets in the future based on hospitals, clinics, and national health repositories. This would enhance the generalizability of the model to various populations, age groups and ethnicities.

- State-of-the-art Machine learning models.

The existing system is based on the classical classification algorithms. Future directions may include deep learning neural networks including convolutional neural networks (CNNs) to analyse ECG signals or recurrent neural networks (RNNs) to analyse time-series patient data. A combination of multiple models in ensemble learning can also be more accurate at prediction.

- Engineering of Features and Expansion of Risk Factors.

Other variables such as genetic predisposition, lifestyle habits (diet, exercise, smoking, alcohol consumption) and psycho-social (stress levels, socioeconomics) variables could be added to get a more holistic picture of risk-assessment.

7.2 Clinical Integration

- IoT and Real-Time Monitoring.

Integrating wearables and IoT sensors will enable the continuous monitoring of vital indicators of patients, including heart rate, blood pressure, and oxygen concentration. This would facilitate active risk forecasting and early warnings of the patients and health care providers.

- Electronic Health Records (EHRs) Interoperability.

The next generation DSS systems must be developed to be compatible with the hospital information systems and EHR operating systems. It would enable clinicians to obtain the predictions directly in their workflow, eliminating the need to duplicate the efforts and enhance efficiency.

- Clinical Trials and Clinical validation.

Reliability should be achieved by conducting intense testing of the system in clinical settings. Controlled trials and pilot studies can be used to support predictive accuracy, usability and patient outcome.

7.3 User-Centric Design and Usability.

- Improved Visualization Methods.

Even more sophisticated visualization tools might be included in future dashboards, including interactive heatmaps, or patient trajectory graphs and predictive trend lines. These tools would support the clinicians in the process of interpreting complex data more intuitively.

- Personalized Interfaces

They should have customizable dashboards that are specific to various user groups (e.g. cardiologists, general practitioners, public health officials).

- Mobile and Cloud Deployment

This can be achieved by developing mobile applications and cloud-based platforms that will enable healthcare professionals to access the DSS remotely to facilitate telemedicine and delivery of healthcare to rural areas.

7.4 Broader Healthcare Impact

- **Educative Healthcare Campaigns.**

The information provided by the DSS might be used to create specific public health programs, targeting high-risk groups determined by the system. This would help to maintain preventive measures and minimize the load of cardiovascular diseases.

- **Cost-Benefit Analysis**

The economic outcomes of the application of such DSS tools in healthcare systems should be evaluated in the future, and the possible savings achieved by fewer hospital admissions and emergency intervention could be considered.

- **Ethics or Privacy Concerns.**

The privacy of patient data and healthcare regulations (including the HIPAA or GDPR) will be vital as the system is expanded. The future research should be directed at the investigation of safe data-sharing models and ethical AI practices.

8.0 CONCLUSION

The creation of the Heart Disease Risk Prediction Decision Support System (DSS) is a major milestone towards the inclusion of AI and predictive analytics in healthcare decision-making. The initiation of this project was informed by the fact that cardiovascular diseases are still among the leading causes of death across the world, and that manual screening methods are frequently ineffective and prone to errors, besides being incapable of screening a significant number of patient records within a short period of time. Through the use of pre-formed datasets, including Kaggle Heart Disease Dataset, the system shows how machine learning systems could be used to transform unstructured medical data into useful information that can then aid clinicians in the earlier and more stable detection of high-risk patients.

The project effectively demonstrates how predictive models may be integrated into a user-friendly interface through the methodology chosen including the acquisition of data, the creation of a data dictionary, the model analysis with models, and the creation of a dashboard. The interactive charts, KPI cards, and slicers included in the Power BI dashboard enable medical workers to have a visual understanding of the risk criteria that are associated with patients in a straightforward and easily understandable manner. This does not only lessen the cognitive load of manual analysis, but also improves the situational awareness, allowing a higher rate of prioritization of patients in need of urgent care. The usability factors of the system also provide the fact that the medical personnel will be able to understand the results intuitively, thus balancing the challenge between complicated analytics and the applicability to a real clinical setting.

This project has made several significant contributions as shown by its findings. To begin with, the prototype of the DSS proves that the integration of demographic, lifestyle, and clinical indicators into a single analytical tool implies more accurate risk prediction as opposed to the separate manual evaluation. Second, the system highlights the promise of AI-powered solutions to decrease the cost of healthcare, as it will reduce cases of late diagnosis and avoidable treatment delays. Third, the project demonstrates how decision support systems can enable healthcare professionals to make evidence-based and individualized decisions, which eventually lead to better patient outcomes and broaden their impact on the general population's health.

In addition to the scope of its particular subject, the project also adds to the increasing literature on the necessity of digital transformation in healthcare. It shows the ways in which predictive analytics could be used to reveal concealed trends in cardiovascular health, minimize human error and foster uniformity in assessing patients. Notably, DSS prototype can be used as a starting point in the future upgrades, which can include addition of real-time monitoring tools, increasing risk factors to cover lifestyle and genetic information, and integration with electronic health record tools. These guidelines are leading in the direction of a future where the decision support systems are no longer research prototypes, but rather research studies.

To sum up, the given project illustrates that the combination of machine learning, data visualization, and decision support framework can greatly enhance the process of early identification and treatment of heart disease. Despite the limitations that are especially important in the context of diversity of the datasets as well as real-life validation, the DSS prototype is a vivid demonstration of a proof of concept that proves that predictive analytics can revolutionize preventive cardiology. Such systems can transform the way healthcare is delivered, enhance patient care, and become part of the global initiative to decrease the burden of cardiovascular diseases by minimizing the diagnostic delays and improving the accuracy of screening and supporting evidence-based interventions.

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